Automated Data Analysis Executive Report

Objective: Predict presence of heart disease based on clinical measurements

1. Validation Confidence Score: 99%

- Sample size adequacy verified: 303 total instances (n=61 test split).
- Overfitting check passed: generalisation gap is within nominal bound (13.1% <= 15%).
- Empirical validation: 0 of 3 hypothesis tests passed statistical significance (p < 0.05).

2. Predictive Modeling (RandomForestClassifier)

Accuracy: 0.852 | F1 Score: 0.849

3. Strategic Recommendations

* Clinical Metric Prioritization: Prioritize diagnostic review of top influential biometric measurements in triage workflows. [High (Accelerates clinical turnaround)]
* Secondary Verification for Borderline Cases: Ensure cases falling within middle probability quartiles receive dual-practitioner review. [High]